SQL FOR DATA ANALYSIS · LESSON 15 OF 15

End-to-End SQL Case Study & Interview Practice

Put everything together. Work through a complete analysis of the retail business (data quality, KPIs, trends, customer segmentation, retention, product and discount analysis) using joins, CTEs and window functions, then practise 25 classic SQL interview questions with model answers.

SQL class 5 (end-to-end case study) · 2026 · Advanced · 180 min

By the end of this lesson you can

- Run a structured SQL analysis: understand the data, check quality, calculate KPIs, find drivers, and recommend actions
- Combine joins, CTEs, aggregation and window functions in multi-step queries
- Build customer segments (RFM), cohort and retention views in SQL
- Answer common SQL interview questions with clear, correct queries
- Explain results in business language

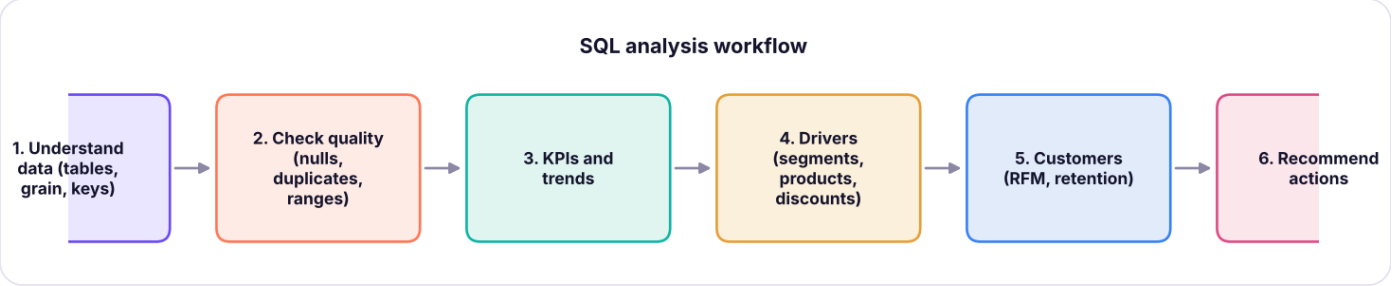
Datasets used

`superstore_sales.csv` `customers.csv` `products.csv`

The business problem

CASE STUDY **Retail growth review**

You are the data analyst at an Indian retailer selling Furniture, Office Supplies and Technology through four regions. Sales grew from 2023 to 2025, but the CFO suspects profit is not keeping pace. Management wants answers to five questions: **(1)** Is the data trustworthy? **(2)** How are sales and profit really trending? **(3)** Where are we losing money and why? **(4)** Who are our best customers and are they staying? **(5)** What should we do next quarter?



A repeatable SQL analysis workflow.

Step 1: understand the data

SQL

```sql
SELECT 'orders'       AS tbl, COUNT(*) AS n, MIN(order_date) AS first_date, MAX(order_date)
AS last_date FROM orders
UNION ALL SELECT 'order_items', COUNT(*), NULL, NULL FROM order_items
UNION ALL SELECT 'customers',   COUNT(*), NULL, NULL FROM customers
UNION ALL SELECT 'products',    COUNT(*), NULL, NULL FROM products;
```

RESULT

tbl	n	first_date	last_date
orders	3200	2023-01-01	2025-12-31
order_items	6323	NULL	NULL
customers	800	NULL	NULL
products	78	NULL	NULL

4 rows × 4 columns

Grain: `order_items` has one row per product line in an order; `orders` one row per order; `customers` one row per customer. KPIs like "orders" must use `COUNT(DISTINCT order_id)`, and "average order value" needs order-level totals (Lesson 10).

Step 2: data-quality checks

Always check before analysing. A compact data-quality report:

SQL

```sql
SELECT
    (SELECT COUNT(*) FROM order_items WHERE sales IS NULL OR profit IS NULL OR quantity IS
NULL)       AS null_measures,
    (SELECT COUNT(*) FROM order_items WHERE sales ⩽ 0 OR quantity ⩽ 0)
AS non_positive,
    (SELECT COUNT(*) FROM order_items WHERE discount < 0 OR discount > 1)
AS bad_discount,
    (SELECT COUNT(*) FROM orders WHERE ship_date < order_date)
AS ship_before_order,
    (SELECT COUNT(*) FROM order_items oi LEFT JOIN orders o ON o.order_id = oi.order_id
WHERE o.order_id IS NULL) AS orphan_items,
    (SELECT COUNT(*) FROM orders o LEFT JOIN customers c ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL) AS orders_without_customer,
    (SELECT COUNT(*) - COUNT(DISTINCT item_id) FROM order_items)
AS duplicate_item_ids;
```

RESULT

null_measures	non_positive	bad_discount	ship_before_order	orphan_items	orders_wi
0	0	0	0	0	0

1 rows × 7 columns

All zeros: the data is clean, so the numbers can be trusted. In real projects every non-zero here becomes a ticket to fix or a documented exclusion. Also check **business-rule** outliers:

SQL

```sql
SELECT ROUND(MIN(sales), 0) AS min_sales, ROUND(MAX(sales), 0) AS max_sales,
       ROUND(AVG(sales), 0) AS avg_sales,
       MIN(discount) AS min_discount, MAX(discount) AS max_discount,
       MIN(quantity) AS min_qty, MAX(quantity) AS max_qty
FROM order_items;
```

RESULT

min_sales	max_sales	avg_sales	min_discount	max_discount	min_qty	max_qty
42	619,290	31,854	0	0.4	1	7

1 rows × 7 columns

Step 3: KPIs and trends

SQL

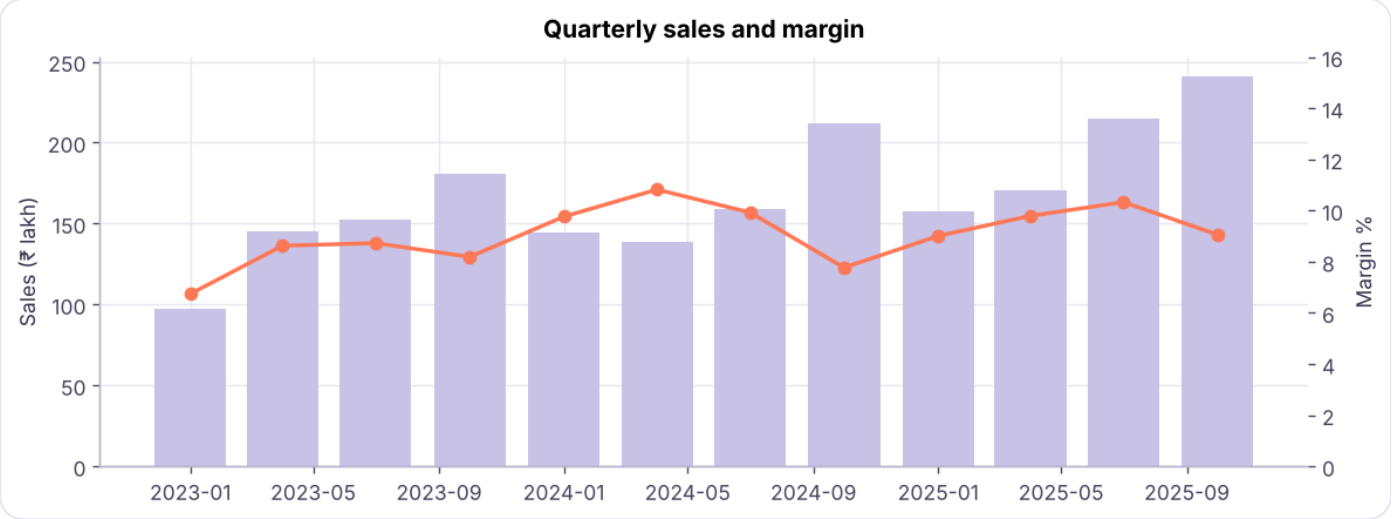
```sql
WITH yearly AS (
    SELECT YEAR(o.order_date)              AS yr,
           COUNT(DISTINCT o.order_id)      AS orders,
           COUNT(DISTINCT o.customer_id)   AS customers,
           SUM(oi.sales)                   AS sales,
           SUM(oi.profit)                  AS profit
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY YEAR(o.order_date)
)
SELECT yr, orders, customers,
       ROUND(sales, 0)                                          AS sales,
       ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY yr)) / LAG(sales) OVER (ORDER BY
yr), 1)   AS sales_growth_pct,
       ROUND(profit, 0)                                         AS profit,
       ROUND(100.0 * (profit - LAG(profit) OVER (ORDER BY yr)) / LAG(profit) OVER (ORDER BY
yr), 1) AS profit_growth_pct,
       ROUND(100.0 * profit / sales, 1)                         AS margin_pct,
       ROUND(sales / orders, 0)                                 AS avg_order_value
FROM yearly
ORDER BY yr;
```

RESULT

yr	orders	customers	sales	sales_growth_pct	profit	profit_growth_pc
2023	883	540	57,640,217	NULL	4,725,378	NULL
2024	1085	590	65,396,556	13.5	6,135,992	29.9
2025	1232	623	78,376,797	19.8	7,490,793	22.1

3 rows × 9 columns

Finding 1. Sales grew 13.5% in 2024 and 19.8% in 2025, and profit grew even faster (29.9% and 22.1%), so the overall margin edged up from 8.2% to 9.6%. The growth is real, but a 9.6% margin is low compared with the 17–18% earned on lines sold **without** a discount (you will see this in Step 4). The CFO's concern is about the *quality* of profit: let's look at where the money is made and lost.



Quarterly sales and profit margin from the same data. Sales climb steadily and peak each Q4 (festive season); the margin moves in a narrow band below 12%.

Step 4: where do we lose money?

SQL

```sql
SELECT p.category, p.sub_category,
       COUNT(*)                                          AS lines,
       ROUND(SUM(oi.sales), 0)                           AS sales,
       ROUND(SUM(oi.profit), 0)                          AS profit,
       ROUND(100.0 * SUM(oi.profit) / SUM(oi.sales), 1)  AS margin_pct,
       ROUND(AVG(oi.discount) * 100, 1)                  AS avg_discount_pct
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category, p.sub_category
ORDER BY profit;
```

RESULT

category	sub_category	lines	sales	profit	margin_pct	avg_discount_pc
Furniture	Tables	505	14,046,224	-816,042	-5.8	17.5
Furniture	Bookcases	477	11,274,849	-422,786	-3.7	18.3
Office Supplies	Art	462	245,985	36,526	14.8	7.6
Office Supplies	Binders	494	401,219	61,074	15.2	7.4
Office Supplies	Paper	559	493,232	75,040	15.2	7.2
Furniture	Furnishings	442	1,429,307	78,632	5.5	7.4
Furniture	Chairs	493	8,091,474	342,043	4.2	8.3
Technology	Accessories	445	3,178,930	369,554	11.6	7.5
Office Supplies	Storage	507	2,659,340	397,228	14.9	7.6
Office Supplies	Appliances	492	6,333,529	944,927	14.9	7.5
Technology	Phones	488	31,919,223	3,663,117	11.5	7.1
Technology	Machines	491	42,590,335	5,186,711	12.2	6.7

13 rows × 7 columns (first 12 shown)

Finding 2. Tables and Bookcases lose money, and they also carry the heaviest average discounts. Is discounting the cause? Compare margin by discount level and by whether the line is Furniture:

SQL

```
SELECT CASE WHEN p.category = 'Furniture' THEN 'Furniture' ELSE 'Other' END AS group_name,
       CASE WHEN oi.discount = 0 THEN '0%' WHEN oi.discount ⩽ 0.2 THEN '1-20%' ELSE '>20%'
END AS discount_band,
       COUNT(*)                                          AS lines,
       ROUND(SUM(oi.profit), 0)                          AS profit,
       ROUND(100.0 * SUM(oi.profit) / SUM(oi.sales), 1)  AS margin_pct
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY CASE WHEN p.category = 'Furniture' THEN 'Furniture' ELSE 'Other' END,
         CASE WHEN oi.discount = 0 THEN '0%' WHEN oi.discount ⩽ 0.2 THEN '1-20%' ELSE
'>20%' END
ORDER BY group_name, discount_band;
```

RESULT

group_name	discount_band	lines	profit	margin_pct
Furniture	0%	850	1,861,761	11.7
Furniture	1–20%	509	-317,393	-3.9
Furniture	>20%	558	-2,362,521	-21.9
Other	0%	2630	19,857,203	18.4
Other	1–20%	1434	1,014,534	2.0
Other	>20%	342	-1,701,419	-18.9

6 rows × 5 columns

Finding 3. In every group, discounts above 20% destroy margin. Furniture is hit hardest because it is discounted more often. What would profit be if lines discounted above 20% had been sold at a 20% discount instead? A rough "what-if":

SQL

```
WITH x AS (
    SELECT oi.sales, oi.profit, oi.discount,
           oi.sales / (1 - oi.discount)           AS list_value,
           (oi.sales / (1 - oi.discount)) * 0.8   AS sales_at_20,
           oi.sales - oi.profit                   AS cost
    FROM order_items oi WHERE oi.discount > 0.2
)
SELECT COUNT(*)                                  AS lines_over_20pct,
       ROUND(SUM(profit), 0)                     AS actual_profit,
       ROUND(SUM(sales_at_20 - cost), 0)         AS profit_if_capped_at_20pct,
       ROUND(SUM(sales_at_20 - cost) - SUM(profit), 0) AS potential_gain
FROM x;
```

RESULT

lines_over_20pct	actual_profit	profit_if_capped_at_20pct	potential_gain
900	-4,063,940	-422,474	3,641,466

1 rows × 4 columns

(This ignores that a lower discount might lose some of those sales. It is an upper bound to size the opportunity, and it must be labelled as such for management.)

Step 5: customers

Top customers and concentration

SQL

```
WITH c AS (
    SELECT o.customer_id, SUM(oi.sales) AS sales, SUM(oi.profit) AS profit, COUNT(DISTINCT
o.order_id) AS orders
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY o.customer_id
), r AS (
    SELECT *, NTILE(10) OVER (ORDER BY sales DESC) AS decile FROM c
)
SELECT decile,
       COUNT(*)                                      AS customers,
       ROUND(100.0 * SUM(sales) / SUM(SUM(sales)) OVER (), 1) AS pct_of_sales,
       ROUND(100.0 * SUM(profit) / SUM(SUM(profit)) OVER (), 1) AS pct_of_profit
FROM r
GROUP BY decile
ORDER BY decile;
```

RESULT

decile	customers	pct_of_sales	pct_of_profit
1	79	28.8	34.1
2	79	18.7	21.0
3	79	14.6	14.3
4	79	11.3	10.4
5	79	9.0	8.8
6	78	6.7	5.3
7	78	5.2	2.5
8	78	3.4	2.2
9	78	1.9	1.1
10	78	0.5	0.3

10 rows × 4 columns

RFM segmentation

RFM scores each customer by **R**ecency (days since last order), **F**requency (number of orders) and **M**onetary value (total sales). The reference date is the day after the last order in the data.

SQL

```sql
WITH base AS (
    SELECT o.customer_id,
           DATEDIFF(day, MAX(o.order_date), '2026-01-01')  AS recency_days,
           COUNT(DISTINCT o.order_id)                      AS frequency,
           SUM(oi.sales)                                   AS monetary
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY o.customer_id
), scored AS (
    SELECT *,
           NTILE(4) OVER (ORDER BY recency_days DESC) AS r,        -- 4 = most recent
           NTILE(4) OVER (ORDER BY frequency)         AS f,
           NTILE(4) OVER (ORDER BY monetary)          AS m
    FROM base
)
SELECT customer_id, recency_days, frequency, ROUND(monetary, 0) AS monetary, r, f, m,
       CASE WHEN r ⩾ 3 AND f ⩾ 3 AND m ⩾ 3 THEN 'Champions'
            WHEN r ⩾ 3 AND f ⩾ 2             THEN 'Loyal'
            WHEN r ⩽ 2 AND f ⩾ 3 AND m ⩾ 3  THEN 'At risk (high value)'
            WHEN r = 1                       THEN 'Lost / hibernating'
            ELSE 'Potential' END AS segment
FROM scored
ORDER BY monetary DESC;
```

RESULT

customer_id	recency_days	frequency	monetary	r	f	m	segment
C0741	441	6	1,204,439	1	4	4	At risk (high value)
C0092	11	8	1,140,540	4	4	4	Champions
C0306	278	7	1,126,867	2	4	4	At risk (high value)
C0781	130	5	1,123,527	2	3	4	At risk (high value)
C0443	92	7	1,113,058	3	4	4	Champions
C0720	12	6	1,100,213	4	4	4	Champions
C0493	514	6	1,004,382	1	4	4	At risk (high value)
C0028	266	5	1,000,189	2	3	4	At risk (high value)
C0630	331	5	1,000,175	1	3	4	At risk (high value)
C0262	70	6	952,738	3	4	4	Champions

785 rows × 8 columns (first 10 shown)

SQL

```
WITH base AS (
    SELECT o.customer_id, DATEDIFF(day, MAX(o.order_date), '2026-01-01') AS recency_days,
           COUNT(DISTINCT o.order_id) AS frequency, SUM(oi.sales) AS monetary
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id GROUP BY o.customer_id
), scored AS (
    SELECT *, NTILE(4) OVER (ORDER BY recency_days DESC) AS r, NTILE(4) OVER (ORDER BY
frequency) AS f, NTILE(4) OVER (ORDER BY monetary) AS m FROM base
), seg AS (
    SELECT *, CASE WHEN r ⩾ 3 AND f ⩾ 3 AND m ⩾ 3 THEN 'Champions' WHEN r ⩾ 3 AND f ⩾
2 THEN 'Loyal'
                   WHEN r ⩽ 2 AND f ⩾ 3 AND m ⩾ 3 THEN 'At risk (high value)' WHEN r = 1
THEN 'Lost / hibernating' ELSE 'Potential' END AS segment FROM scored
)
SELECT segment, COUNT(*) AS customers, ROUND(SUM(monetary), 0) AS sales,
       ROUND(100.0 * SUM(monetary) / SUM(SUM(monetary)) OVER (), 1) AS pct_of_sales
FROM seg GROUP BY segment ORDER BY sales DESC;
```

RESULT

segment	customers	sales	pct_of_sales
Champions	178	80,496,072	40.0
At risk (high value)	92	42,616,303	21.2
Potential	198	29,644,898	14.7
Lost / hibernating	169	24,817,833	12.3
Loyal	148	23,838,464	11.8

5 rows × 4 columns

The "At risk (high value)" group is the first win: high-spending customers who have gone quiet deserve a personal outreach before they churn. The same RFM idea is implemented with pandas in the ML track.

Retention and cohorts

A **cohort** groups customers by the year of their first order. **Retention** shows how many come back in later years.

SQL

```sql
WITH first_order AS (
    SELECT customer_id, YEAR(MIN(order_date)) AS cohort_year FROM orders GROUP BY
customer_id
), activity AS (
    SELECT DISTINCT customer_id, YEAR(order_date) AS active_year FROM orders
)
SELECT f.cohort_year,
       COUNT(DISTINCT f.customer_id)                                          AS
cohort_size,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year     THEN a.customer_id END)
AS y0,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 1 THEN a.customer_id END)
AS y1,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 2 THEN a.customer_id END)
AS y2
FROM first_order f
JOIN activity a ON a.customer_id = f.customer_id
GROUP BY f.cohort_year
ORDER BY f.cohort_year;
```

RESULT

cohort_year	cohort_size	y0	y1	y2
2023	540	540	397	415
2024	193	193	156	0
2025	52	52	0	0

3 rows × 5 columns

Divide y1 by `cohort_size` to get year-1 retention. About 74% of the 2023 cohort (397 of 540) ordered again in 2024, and some who skipped a year returned in 2025: this is a business with sticky customers, so reducing discounts is likely safer than the CFO fears.

Step 6: recommendations

KEY TAKEAWAYS **Findings and actions**

1. **Sales and profit are growing, but the margin is only 9.6%**, far below the 17–18% earned on undiscounted lines.
2. **Technology produces nearly all the profit**; **Furniture is net loss-making** because Tables and Bookcases are sold with heavy discounts.
3. **Discounts above 20% lose money in every category.** Cap discounts at 20% (the what-if sizes the upper-bound upside at about ₹36 lakh over three years; the real gain will be lower if some customers buy less), test in two regions with an A/B design.
4. **Customer base is sticky** (strong year-on-year retention), so price discipline is lower-risk than assumed.
5. **Re-engage "At risk (high value)" customers** with a targeted offer instead of blanket discounts.
6. Build a **monthly SQL view/dashboard** with these KPIs, with data-quality checks as a first step.

SQL interview practice: 25 questions with answers

Run these on the retail database. Each answer is a runnable query.

INTERVIEW CORNER **Q1. Second-highest sales line.**

SQL

```sql
SELECT MAX(sales) AS second_highest FROM order_items WHERE sales < (SELECT MAX(sales)
FROM order_items);
```

RESULT

second_highest
551,530

1 rows × 1 columns

INTERVIEW CORNER

Q2. Customers with at least 8 orders.

SQL

```sql
SELECT customer_id, COUNT(*) AS orders FROM orders GROUP BY customer_id HAVING
COUNT(*) ⩾ 8 ORDER BY orders DESC;
```

RESULT

customer_id	orders
C0219	10
C0396	10
C0291	9
C0664	9
C0110	9
C0384	9
C0596	9
C0419	9
C0271	9
C0527	9
C0719	9
C0069	9

41 rows × 2 columns (first 12 shown)

INTERVIEW CORNER

Q3. Duplicate customer names (possible duplicate records).

SQL

```
SELECT customer_name, COUNT(*) AS n FROM customers GROUP BY customer_name HAVING
COUNT(*) > 1 ORDER BY n DESC;
```

RESULT

customer_name	n
Sneha Joshi	6
Ananya Gupta	5
Rahul Shah	5
Asha Rao	5
Ananya Iyer	5
Saanvi Khan	5
Rohan Gupta	5
Reyansh Shah	4
Simran Khan	4
Vikram Kulkarni	4
Arjun Sharma	4
Nikhil Das	4

238 rows × 2 columns (first 12 shown)

INTERVIEW CORNER

Q4. Highest-selling product in each category.

SQL

```
WITH t AS (SELECT p.category, p.product_name, SUM(oi.sales) AS sales FROM order_items
oi JOIN products p ON p.product_id = oi.product_id GROUP BY p.category,
p.product_name),
     r AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY category ORDER BY sales DESC) AS
rn FROM t)
SELECT category, product_name, ROUND(sales, 0) AS sales FROM r WHERE rn = 1;
```

RESULT

category	product_name	sales
Furniture	Crest Table 572	3,122,680
Technology	Orbit Copier 746	16,026,341
Office Supplies	Crest Appliance 402	1,909,628

3 rows × 3 columns

INTERVIEW CORNER

Q5. Month-over-month growth for 2025.

SQL

```sql
WITH m AS (SELECT MONTH(order_date) AS mth, SUM(sales) AS sales FROM superstore WHERE YEAR(order_date) = 2025 GROUP BY MONTH(order_date))
SELECT mth, ROUND(sales, 0) AS sales, ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY mth)) / LAG(sales) OVER (ORDER BY mth), 1) AS mom_pct FROM m ORDER BY mth;
```

RESULT

mth	sales	mom_pct
1	5,357,761	NULL
2	3,879,338	-27.6
3	6,512,645	67.9
4	6,699,805	2.9
5	6,421,274	-4.2
6	3,968,572	-38.2
7	7,190,087	81.2
8	5,802,184	-19.3
9	8,468,060	45.9
10	6,140,290	-27.5
11	10,284,516	67.5
12	7,652,267	-25.6

12 rows × 3 columns

INTERVIEW CORNER

Q6. Running total of sales by month in 2025.

SQL

```sql
WITH m AS (SELECT MONTH(order_date) AS mth, SUM(sales) AS sales FROM superstore WHERE YEAR(order_date) = 2025 GROUP BY MONTH(order_date))
SELECT mth, ROUND(sales, 0) AS sales, ROUND(SUM(sales) OVER (ORDER BY mth ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW), 0) AS running_total FROM m ORDER BY mth;
```

RESULT

mth	sales	running_total
1	5,357,761	5,357,761
2	3,879,338	9,237,099
3	6,512,645	15,749,743
4	6,699,805	22,449,548
5	6,421,274	28,870,822
6	3,968,572	32,839,394
7	7,190,087	40,029,481
8	5,802,184	45,831,665
9	8,468,060	54,299,725
10	6,140,290	60,440,015
11	10,284,516	70,724,530
12	7,652,267	78,376,797

12 rows × 3 columns

INTERVIEW CORNER

Q7. Customers who ordered in every year (2023, 2024 and 2025).

SQL

```sql
SELECT COUNT(*) AS customers_all_three_years
FROM (SELECT customer_id FROM orders GROUP BY customer_id HAVING COUNT(DISTINCT
YEAR(order_date)) = 3) AS t;
```

RESULT

customers_all_three_years
311

1 rows × 1 columns

INTERVIEW CORNER

Q8. Employees earning more than their department's average (HR data).

SQL

```sql
SELECT COUNT(*) AS above_dept_avg
FROM (SELECT emp_id, monthly_income, AVG(monthly_income) OVER (PARTITION BY
department) AS avg_income FROM employees) AS t
WHERE monthly_income > avg_income;
```

RESULT

above_dept_avg
638

1 rows × 1 columns

INTERVIEW CORNER

Q9. Delete duplicates keeping the lowest id (pattern).

SQL

```sql
WITH d AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_name, city ORDER BY
customer_id) AS rn FROM dbo.customers)
DELETE FROM d WHERE rn > 1;
```

INTERVIEW CORNER

Q10. Customers with no orders in the last 6 months of 2025 but orders before that (churn candidates).

SQL

```sql
SELECT COUNT(*) AS churn_candidates
FROM (SELECT customer_id FROM orders WHERE order_date < '2025-07-01' GROUP BY
customer_id
      EXCEPT
      SELECT customer_id FROM orders WHERE order_date ⩾ '2025-07-01') AS t;
```

RESULT

churn_candidates
310

1 rows × 1 columns

INTERVIEW CORNER

Q11. Percentage of orders shipped late (more than 5 days).

SQL

```sql
SELECT ROUND(100.0 * SUM(CASE WHEN DATEDIFF(day, order_date, ship_date) > 5 THEN 1
ELSE 0 END) / COUNT(*), 1) AS pct_late FROM orders;
```

RESULT

pct_late
31.3

1 rows × 1 columns

INTERVIEW CORNER

Q12. Median order-line sales.

SQL

```sql
SELECT DISTINCT PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY sales) OVER () AS
median_sales FROM order_items;
```

RESULT

median_sales
8,430

1 rows × 1 columns

INTERVIEW CORNER

Q13. Categories where more than 30% of lines are loss-making.

SQL

```sql
SELECT p.category, ROUND(100.0 * AVG(CASE WHEN oi.profit < 0 THEN 1.0 ELSE 0 END), 1)
AS pct_loss_lines
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category HAVING AVG(CASE WHEN oi.profit < 0 THEN 1.0 ELSE 0 END) > 0.3;
```

RESULT

category	pct_loss_lines
Furniture	48.9

1 rows × 2 columns

INTERVIEW CORNER

Q14. Cumulative percentage of sales by customer (Pareto): how many customers make 50% of sales?

SQL

```sql
WITH c AS (SELECT customer_id, SUM(sales) AS s FROM order_items oi JOIN orders o ON
o.order_id = oi.order_id GROUP BY customer_id),
     p AS (SELECT s, SUM(s) OVER (ORDER BY s DESC ROWS BETWEEN UNBOUNDED PRECEDING AND
CURRENT ROW) * 1.0 / SUM(s) OVER () AS cum FROM c)
SELECT COUNT(*) AS customers_for_half_of_sales, (SELECT COUNT(*) FROM c) AS
total_customers FROM p WHERE cum ⩽ 0.5;
```

RESULT

customers_for_half_of_sales	total_customers
170	785

1 rows × 2 columns

INTERVIEW CORNER

Q15. Swap two values in a column (pattern).

```sql
UPDATE dbo.staff SET gender = CASE gender WHEN 'M' THEN 'F' WHEN 'F' THEN 'M' ELSE
gender END;
```

INTERVIEW CORNER

Q16. Find consecutive order days (streaks) for a customer: see Lesson 12 (gaps and islands).

INTERVIEW CORNER

Q17. Average order value per segment.

```sql
SELECT segment, ROUND(AVG(t.total), 0) AS aov
FROM (SELECT c.segment, o.order_id, SUM(oi.sales) AS total FROM orders o JOIN
customers c ON c.customer_id = o.customer_id JOIN order_items oi ON oi.order_id =
o.order_id GROUP BY c.segment, o.order_id) AS t
GROUP BY segment;
```

RESULT

segment	aov
Consumer	63,451
Corporate	61,899
Home Office	63,237

3 rows × 2 columns

INTERVIEW CORNER

Q18. Products priced above the average of their sub-category.

SQL

```sql
SELECT COUNT(*) AS products_above_subcat_avg
FROM (SELECT product_id, list_price, AVG(list_price) OVER (PARTITION BY sub_category)
AS avg_price FROM products) AS t WHERE list_price > avg_price;
```

RESULT

products_above_subcat_avg
36

1 rows × 1 columns

INTERVIEW CORNER

Q19. Rows in A not in B without NOT IN trap: use NOT EXISTS (Lesson 10).

INTERVIEW CORNER

Q20. Pivot category sales into columns by year.

SQL

```sql
SELECT category,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2023 THEN sales ELSE 0 END), 0) AS
y2023,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2024 THEN sales ELSE 0 END), 0) AS
y2024,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2025 THEN sales ELSE 0 END), 0) AS y2025
FROM superstore GROUP BY category ORDER BY category;
```

RESULT

category	y2023	y2024	y2025
Furniture	9,952,975	12,128,633	12,760,246
Office Supplies	2,611,874	3,550,109	3,971,321
Technology	45,075,369	49,717,814	61,645,230

3 rows × 4 columns

INTERVIEW CORNER

Q21. First and last order date and the number of days between for each customer (top 3 longest tenures).

SQL

```sql
SELECT TOP 3 customer_id, MIN(order_date) AS first_order, MAX(order_date) AS
last_order, DATEDIFF(day, MIN(order_date), MAX(order_date)) AS days_active
FROM orders GROUP BY customer_id ORDER BY days_active DESC;
```

RESULT

customer_id	first_order	last_order	days_active
C0079	2023-01-08	2025-12-27	1084
C0192	2023-01-24	2025-12-31	1072
C0070	2023-01-09	2025-12-13	1069

3 rows × 4 columns

INTERVIEW CORNER

Q22. Which month has the highest sales in each year?

SQL

```sql
WITH m AS (SELECT YEAR(order_date) AS yr, MONTH(order_date) AS mth, SUM(sales) AS
sales FROM superstore GROUP BY YEAR(order_date), MONTH(order_date)),
     r AS (SELECT *, RANK() OVER (PARTITION BY yr ORDER BY sales DESC) AS rnk FROM m)
SELECT yr, mth, ROUND(sales, 0) AS sales FROM r WHERE rnk = 1 ORDER BY yr;
```

RESULT

yr	mth	sales
2023	12	7,152,348
2024	11	7,284,664
2025	11	10,284,516

3 rows × 3 columns

INTERVIEW CORNER

Q23. How do you find the number of rows in each table of a database?

SQL

```sql
SELECT t.name AS table_name, SUM(p.rows) AS row_count
FROM sys.tables t JOIN sys.partitions p ON p.object_id = t.object_id AND p.index_id IN
(0, 1)
GROUP BY t.name ORDER BY row_count DESC;
```

INTERVIEW CORNER

Q24. Explain the difference between RANK() and ROW_NUMBER() and when each is right: Lesson 11.

INTERVIEW CORNER

Q25. How would you explain the result of a query to a non-technical manager?

State the answer first, in one sentence with numbers ("Furniture loses ₹8.2 lakh a year, almost entirely on heavily discounted tables and bookcases"), then give the 1–2 facts that support it, the assumption or limitation, and the action you recommend.

Summary

KEY TAKEAWAYS

- A good analysis follows a workflow: understand the grain, check data quality, calculate KPIs, find drivers, study customers, recommend actions.
- Always check data quality first; label what-if estimates as estimates.
- Combine CTEs, joins and window functions to build RFM segments, concentration (Pareto), cohorts and retention.
- In interviews, state your assumptions, clarify the grain, write clean, formatted SQL, and explain the result in business language.

Quick quiz

6 questions · tap an answer to check it

1. Why must "number of orders" use COUNT(DISTINCT order_id) when joining orders to order_items?

A DISTINCT is faster

B Each order appears once per item, so COUNT(*) would count lines

C order_id is a text column

D It avoids NULLs

Answer: B. The join repeats the order for each of its items.

2. **In RFM, a customer with recency 400 days, frequency 1 and low monetary value is probably:**

A A champion

B Lost or hibernating

C A loyal customer

D A new high-value customer

Answer: B. Long time since the last order, few orders and low spend.

3. **A cohort table shows 2023 cohort size 500, year-1 active 350. Year-1 retention is:**

A 35%

B 70%

C 50%

D 150%

Answer: B. 350 ÷ 500 = 70%.

4. **Which should you do before reporting KPIs?**

A Build a dashboard

B Check data quality: nulls, duplicates, orphans, ranges

C Email the CFO

D Add indexes

Answer: B. Untrusted data produces untrusted numbers.

5. **The what-if "profit if discounts were capped at 20%" is best described as:**

A A forecast

B A guarantee

C An upper-bound estimate that ignores possible lost sales

D A data error

Answer: C. It assumes demand would not change, so it overstates the benefit.

6. **NTILE(10) over customers ordered by sales gives:**

A Ten customers

B Ten equal-sized groups (deciles) of customers

C The top 10% only

D The median

Answer: B. NTILE splits rows into n buckets.

EXERCISE **Capstone: your own analysis**

Using only SQL on the retail database, prepare a one-page analysis answering: (1) Which three cities have the highest profit and which three the lowest? (2) What is the average days to ship by ship mode, and does a slower mode save money (profit per line)? (3) Which product's sales fell the most from 2024 to 2025? (4) Propose two actions with the numbers that support them.

Show solution (worked answer)

Hints: (1) `GROUP BY city` with `ORDER BY profit` ascending and descending using TOP 3 in two queries or `RANK()` in a CTE. (2) Join orders to order_items, `AVG(DATEDIFF(day, order_date, ship_date))` and `SUM(profit)/COUNT(*)` by ship_mode. (3) Pivot product sales by year with conditional aggregation and compute the difference. (4) Example actions: cap Furniture discounts at 20% in two pilot regions; re-engage at-risk high-value customers. Present each action with a baseline number, an expected impact (labelled as an estimate) and a way to measure success.

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